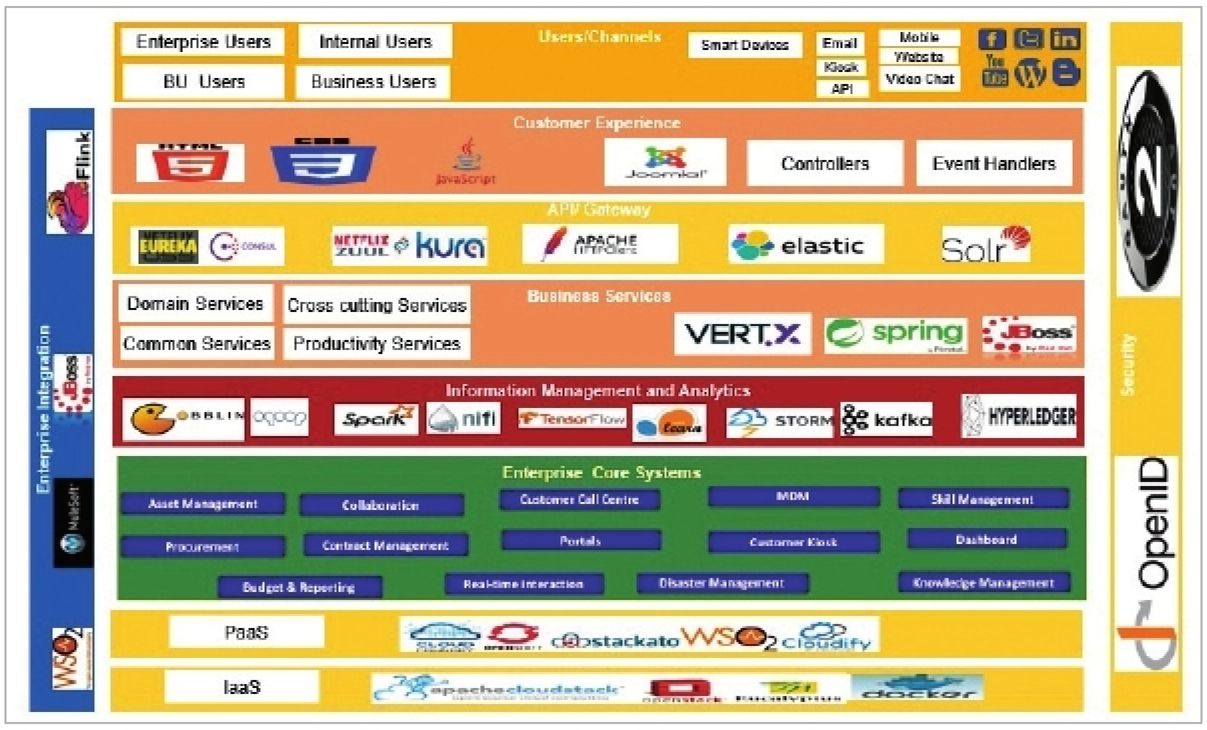

Figure 3: Open source adoption in enterprises

developer participation for the benefit of the community.

API management includes the full API management life cycle, as well as the integration with diverse applications and appliances to support management, monitoring and tracking of all your API activities.

Large complex systems in enterprises follow the API-centric approach when exposing services for integration with multiple third party solutions. The APIs are RESTful, XML based, and stateless services. All APIs are accessible through the HTTPS protocol.

Service discoverability: This is the process of client applications querying the central registry to learn the location of services. Service discoverability ensures that the metadata attached to a service describes the overall purpose of the service and its functionality, which makes the services easily discoverable. A repository of reusable business logic components needs to be maintained and made available to enable service discoverability.

Consul, Apache Zookeeper, Etcd, Netflix Eureka, etc, are the widely used open source service discovery tools.

Platform and database agnostic: Applications must be forward and backward compatible as well as deployable on any technology platform. They should be able to communicate with any data store.

For example, Jenkins is a continuous delivery tool that builds and tests software after every change. Jenkins can deploy new code to production, which helps in no downtime when making the upgraded product available to constituents.

PostGreSQL, MySQL, MongoDB and CouchBase are the widely used open source databases.

Microservices: The design of the business and information services layer of the enterprise application is developed using microservices architecture principles and cross-channel capabilities. Micro Services Architecture (MSA) allows the creation of services that are loosely coupled and have different programming language bases. These are scalable and have a quicker delivery time. MSA enables the development of new enterprise applications as reusable microservices based on open standards, which expose the standard recommended integration interfaces. This ensures independent scaling of individual services within an application with better resource utilisation.

Application life cycle automation tools like Ansible help in tracking, deploying and measuring the changes in and enhancements of microservices.

Mozilla, Jboss, AngularJS and Bootstrap are the frequently used open source development tools today. Other popular open source tools are Joomla, Jetty, OpenSSO, SOLR, Chef, Spring and WordPress.

Deployment architecture: The deployment platform chosen should be one that reduces the overall deployment complexity and supports open source based architectures. The suggestion is to consider using a dynamic component system for Java (e.g., OSGi).

Components can be installed, started, stopped, updated and be uninstalled without bringing down the whole system. This can significantly reduce deployment times.

Integration: The value of a strong integration system lies in its ability to connect to a wide range of backend systems. It provides mediation, transformation, protocol and routing capabilities, and acts as a gateway to integrate with core back-end systems. It also provides aggregation and broker communications. Therefore, it is important to consider an experienced integration solutions vendor who has the ability to connect to other databases, frameworks, applications and endpoints.

It is important to go with open standards and open interfaces while choosing the integration back-end as it should be free from vendor-specific modifications and proprietary hooks. Mule, JBoSS and WS02 are the famous open source integration products that can be used to establish the digital enterprise.

Security: Effective enterprise security is the need of the hour. It ensures that all the components of the cloud infrastructure are guarded against the fast growing security threats. From the application perspective, applications and endpoints need to be protected, and users granted access to applications as per the eligibility configured as part of the RBAC (role-based access control).

Oauth2 and OpenID are a few of the open source tools that can be used for enterprise security.

Customer experience: The customer experience is the sum of all the interactions between an organisation and a customer over the duration of their relationship. A good customer experience means that the individual's experience at all points of contact matches expectations. While building applications, make sure that they